

High-Side Gate Bump and Voltage Utilization of the EPC2361 SPB Building Block

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Summary.

- When the low side turns on, the gate of the held-off high side rises to 0.99 V (LTspice, Q3D parasitics, 75 V, 141 A), above the 0.8 V minimum threshold of EPC2361.
- The bump is the Miller charge of the high side divided onto its gate–source capacitance, $\Delta v_{GS} \approx Q_{GD}/C_{GS} \approx 1.06$ V. It is almost independent of the load current (0.95–0.99 V from 10 to 157 A) and of the bus voltage (0.96, 0.98 and 0.99 V at 48, 60 and 75 V). **Neither lowering the current nor lowering the submodule voltage removes it.**
- The bump lasts about 2 ns above 0.8 V. Even with the high-side threshold lowered to 0.6 V, the simulation shows no channel shoot-through: the high-side charge (output capacitance) and E_{on} are unchanged.
- The EPC90156 evaluation board (same transistor) uses 1 Ω /0.47 Ω gate resistors, 0 V off-state and no gate capacitor, and is rated for 80 V; it has the same bump mechanism.
- Robust fix: an off-state bias of -1 V moves the peak to -0.01 V (0.8 V margin) at a cost of 0.01 % efficiency. -2 V brings the undershoot to -3.26 V, close to the -4 V limit. A 2 nF external C_{GS} only reaches 0.68 V and costs 21 % more E_{on} .
- Voltage utilization: with an overshoot of about 11 V and a battery range of $k = V_{m,max}/V_m = 1.2$, the peak v_{DS} is 101 V for $V_m = 75$ V (above the rating) and 83 V for 60 V. Keeping the peak below 80 V (80 % of the rating, as EPC rates its own board) allows $V_m \leq 57$ V for $k = 1.2$ and ≤ 69 V for a regulated bus. **The voltage limit, not the gate bump, is the reason to reduce the submodule voltage to about 60 V ($N = 20$), at the cost of 0.09 % efficiency and 25 % more transistors.**

1 Observation

In the double-pulse simulation of the building block (two EPC2361 per switch, centre driver, Q3D power loop and gate star, external $R_{g,on}/R_{g,off} = 2/0 \Omega$), the gate of the high side, which is held off through the driver pull-down, rises to about 1 V when the low side turns on (Fig. 1a). The datasheet threshold of EPC2361 is 0.8 V minimum, 1.1 V typical and 2.5 V maximum at 25 °C, and it decreases with temperature (datasheet Fig. 10). A bump of this size therefore has no guaranteed margin.

2 Mechanism

During the low-side turn-on the high-side drain–source voltage rises from about -2 V (reverse conduction) to the bus voltage within 4–6 ns. The displacement current through C_{GD} charges the high-side gate. If the hold-off path could remove this charge instantly, the gate would stay at the off level; it cannot, because its time constants are of the same order as the transition:

$$\tau_{RC} = R_{G,off} C_{iss} \approx 0.8 \Omega \cdot 3.6 \text{ nF} \approx 2.9 \text{ ns}, \quad \tau_L = L_G/R_{G,off} \approx 1.7 \text{ nH}/0.8 \Omega \approx 2.1 \text{ ns}. \quad (1)$$

In the limit of a hold-off path that does not react, the Miller charge is shared between C_{GD} and C_{GS} :

$$\Delta v_{GS} \approx \frac{Q_{GD}}{C_{GS} + C_{ext}} = \frac{3.8 \text{ nC}}{3.59 \text{ nF}} \approx 1.06 \text{ V}, \quad (2)$$

and the simulated peak of 0.99 V is close to this limit. Two consequences follow, both confirmed by the simulations (Fig. 1b, c):

- **Current independence.** Q_{GD} is a property of the voltage swing, not of the current. Over 10–157 A the peak changes by less than 0.05 V for every gate-resistor pair, although the turn-on dv/dt falls from 18 to 12 V/ns.
- **Weak voltage dependence.** C_{GD} of a GaN HEMT is large only at low v_{DS} and drops by orders of magnitude within the first 10–20 V, so most of Q_{GD} is transferred early in the transition. Switching 48, 60 or 75 V moves almost the same Miller charge: 0.96, 0.98 and 0.99 V.

A larger turn-off resistance makes the bump worse (1.10, 1.21 and 1.29 V for 2, 5 and 10 Ω), and a slower turn-on helps little (0.94 V for $R_{g,on} = 10 \Omega$, at 2.5 times E_{on}).

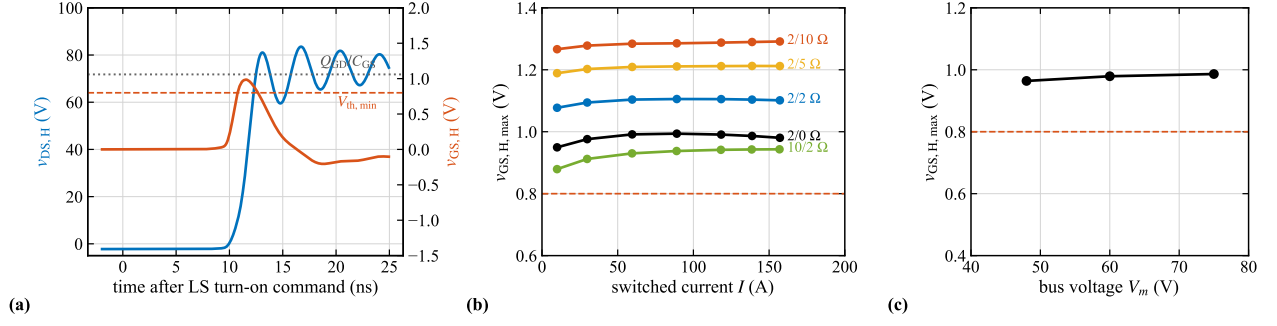


Figure 1: (a) High-side drain–source and gate–source voltage during the low-side turn-on (75 V, 141 A, 2/0 Ω); dashed: $V_{th,min}$, dotted: Q_{GD}/C_{GS} . (b) Peak high-side gate voltage against the switched current for five gate-resistor pairs $R_{g,on}/R_{g,off}$. (c) Peak high-side gate voltage against the bus voltage (141 A, 2/0 Ω).

3 Does the high side turn on?

The bump lasts 1.5–1.9 ns above 0.8 V and about 3 ns above 0.5 V. To test whether it causes conduction, the threshold of both high-side transistors was lowered by 0.3 V (to the datasheet minimum of 0.8 V) and by 0.5 V (0.6 V) with a series voltage in the model. Table 1 shows that the positive high-side charge during the low-side turn-on, which is the charging of its output capacitance, and E_{on} do not increase: the channel does not conduct measurably within the 2 ns. In the model, the bump is therefore not harmful. The margin is nevertheless small for hardware, because the model excludes the package inductance and routing details, the threshold falls at high temperature, and the gate ringing in hardware is usually stronger than simulated. The peak itself is a narrow spike; it varies between 0.99 and 1.09 V with the solver time step.

Table 1: High-side conduction check (75 V, 141 A, 2/0 Ω , 0 V off-state)

High-side threshold	1.1 V (typical)	0.8 V (min.)	0.6 V
Positive high-side charge (left die)	221 nC	219 nC	218 nC
E_{on} of the low side	40.9 μ J	40.6 μ J	40.5 μ J

4 Comparison with the EPC90156 evaluation board

EPC’s half-bridge board for the same transistor (quick-start guide and schematic Rev. 4.0, in Datasheet/) does not use any special measure against the bump (Table 2). It drives one transistor per switch with split outputs, 1 Ω for turn-on and 0.47 Ω for turn-off, a 0/5 V gate drive without negative bias and without a gate–source capacitor. Its input is rated for 80 V, i.e., 80 % of the transistor rating, and all measurements in the guide are at 48 V. Since the bump is set by the device ratio Q_{GD}/C_{GS} , the evaluation board has the same order of bump; EPC accepts it, relying on the short duration shown above. What differs is the voltage derating.

Table 2: Gate drive of the EPC90156 and of the building block

	EPC90156	building block
Transistors per switch	1 $\times$ EPC2361	2 $\times$ EPC2361
Driver	uP1966E, split outputs, 5 V	LT8418, split outputs, 5 V
$R_{g,on} / R_{g,off}$ (external)	1 / 0.47 Ω	2 / 0 Ω
Off-state bias / $C_{GS,ext}$	0 V / none	0 V / none
Hold-off path	one pull-down per transistor	one pull-down shared by two, star with $L_G = 1.71$ nH
Rated input voltage	80 V (measured at 48 V)	75 V nominal
Rated current	40 A	100 A _{rms} (141 A peak)

5 Mitigation

The model was extended by an off-state bias $-V_{neg}$ of the driver (both drivers) and an external capacitor C_{ext} from every gate pad to its source pad, and simulated for $V_{neg} = 0, 1, 2$ V, $C_{ext} = 0, 1, 2$ nF and 48, 60, 75 V

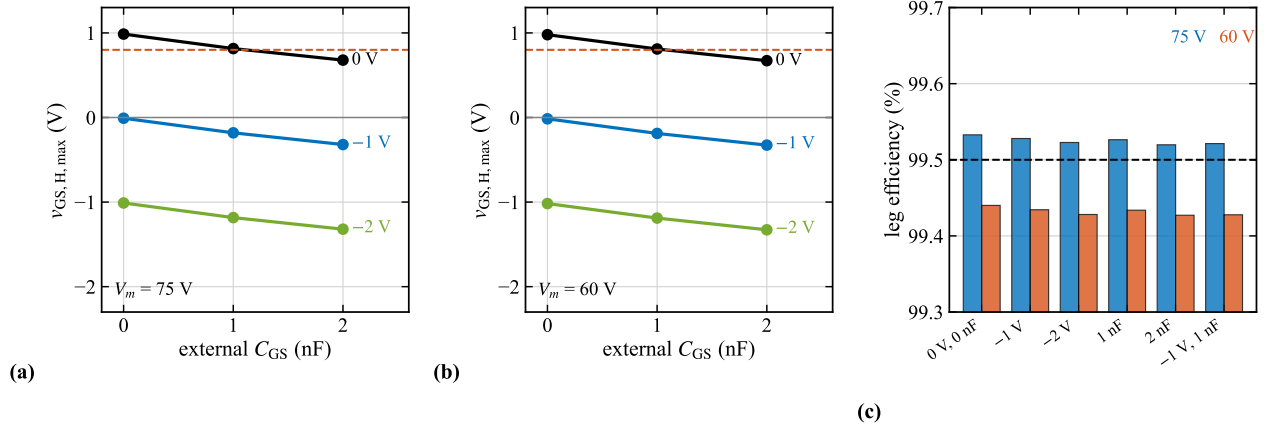


Figure 2: Peak high-side gate voltage against the external gate–source capacitance for off-state biases of 0, –1 and –2 V at (a) 75 V and (b) 60 V (141 A); dashed: $V_{th,min}$. (c) Leg efficiency at 50 kHz for the options at 75 and 60 V.

Table 3: Gate-bump options (141 A; losses per leg at 50 kHz, rated point)

V_m [V]	bias, C_{ext}	$v_{GS,H,max}$ [V]	$v_{GS,H,min}$ [V]	E_{on} [μJ]	E_{off} [μJ]	P_{sw} [W]	P_{dead} [W]	P_{gate} [W]	η [%]
75	0 V, –	0.99	-1.27	40.9	8.0	1.91	0.20	0.03	99.53
75	-1 V, –	-0.01	-2.26	41.8	7.8	1.94	0.29	0.04	99.53
75	-2 V, –	-1.01	-3.26	42.8	7.8	1.98	0.38	0.05	99.52
75	0 V, 1 nF	0.82	-0.97	45.2	7.9	2.08	0.20	0.03	99.53
75	0 V, 2 nF	0.68	-0.79	49.6	7.9	2.25	0.20	0.04	99.52
75	-1 V, 1 nF	-0.18	-1.97	46.3	7.7	2.11	0.29	0.05	99.52
60	0 V, –	0.98	-1.25	29.9	6.1	1.41	0.20	0.03	99.44
60	-1 V, –	-0.02	-2.25	30.8	5.8	1.43	0.29	0.04	99.43
60	-2 V, –	-1.02	-3.25	31.6	5.8	1.46	0.38	0.05	99.43
60	0 V, 1 nF	0.81	-0.95	33.4	6.0	1.54	0.20	0.03	99.43
60	0 V, 2 nF	0.67	-0.77	36.9	6.0	1.67	0.20	0.04	99.43
60	-1 V, 1 nF	-0.19	-1.95	34.3	5.8	1.56	0.29	0.05	99.43
48	0 V, –	0.96	-1.22	21.9	4.7	1.04	0.20	0.03	99.32
48	-1 V, –	-0.03	-2.23	22.6	4.4	1.06	0.29	0.04	99.32
48	-2 V, –	-1.03	-3.23	23.3	4.4	1.08	0.38	0.05	99.31
48	0 V, 1 nF	0.80	-0.93	24.5	4.6	1.14	0.20	0.03	99.32
48	0 V, 2 nF	0.66	-0.75	27.3	4.6	1.25	0.20	0.04	99.31
48	-1 V, 1 nF	-0.20	-1.93	25.3	4.4	1.16	0.29	0.05	99.31

(simulation/bb_spice/BB_bump.cir, 27 cases). The leg loss uses the switching energies of each case, the dead-time loss with the reverse-conduction drop increased by V_{neg} , and the gate-drive loss including C_{ext} (Fig. 2, Table 3).

- **Negative off-state bias** shifts the whole waveform: the peak follows $-V_{neg}$ almost exactly (-0.01 V for -1 V, -1.01 V for -2 V at 75 V). The cost is small: the dead-time loss rises from 0.20 to 0.38 W at -2 V and the turn-off becomes slightly faster, which raises the low-side overshoot from 86.5 to 90.2 V at -2 V. At -2 V the high-side gate undershoots to -3.26 V when the low side turns off, only 0.7 V from the -4 V absolute maximum. **-1 V is the best compromise:** 0.8 V margin to $V_{th,min}$, undershoot -2.3 V.
- **External gate–source capacitor** reduces the bump according to (2) (0.82 and 0.68 V for 1 and 2 nF), but it slows the turn-on of the switching transistor as well: E_{on} rises from 40.9 to 49.6 μJ at 2 nF. Halving the bump would need about 3.6 nF.
- **Combination** -1 V and 1 nF gives -0.18 V.

All options keep the leg efficiency at 75 V between 99.52 and 99.53 %.

6 Voltage utilization of the 100 V transistors

The submodule voltage V_m determines how well the 100 V rating is used. At a fixed inverter power and bus voltage the phase current is fixed (100 A_{rms}), so the power per leg, and per transistor, is proportional to V_m , whereas the conduction

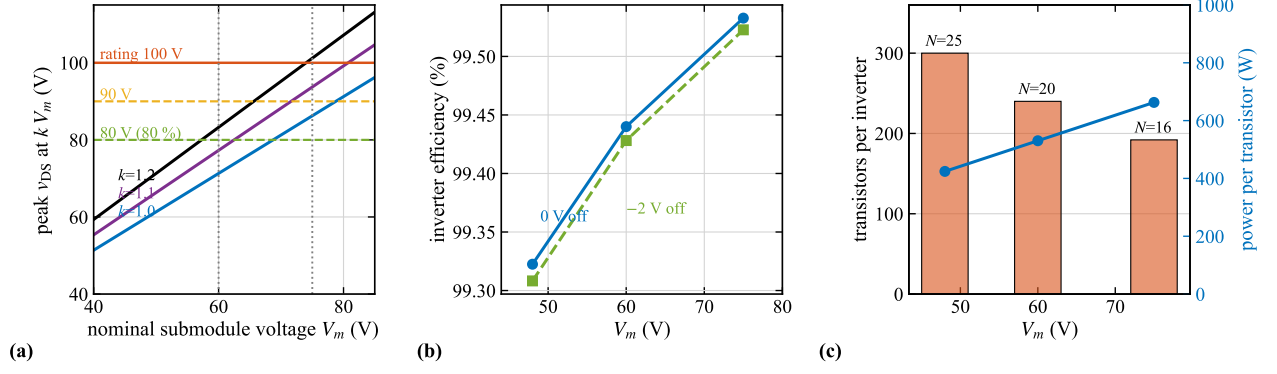


Figure 3: Voltage utilization. (a) Peak drain–source voltage against the nominal submodule voltage for maximum-to-nominal ratios $k = 1.0, 1.1, 1.2$ (overshoot from the 141 A simulations); lines: rating, 90 V and 80 V (80 % derating). (b) Inverter efficiency at the rated point without and with -2 V off-state bias. (c) Number of transistors of the 1200 V, 150 kW inverter and power per transistor ($M = 1.15$).

and copper losses per leg are not. A higher V_m therefore gives fewer submodules and transistors and a higher efficiency (Fig. 3b, c; Table 4). The upper limit is the peak drain–source voltage:

$$\hat{v}_{DS} = k V_m + \Delta v_{os}, \quad k = V_{m,max}/V_m, \quad (3)$$

where k covers the battery-voltage range and the unbalance of the submodules, and the overshoot is $\Delta v_{os} \approx 11.5$ V at 141 A (10.7 V at 60 V; Q3D parasitics, larger at 160 A). For the original assumption of 75 V nominal and 90 V maximum ($k = 1.2$), the peak is 101 V, i.e., above the rating (Fig. 3a). Keeping the peak at or below 80 V, the derating that EPC applies to its own board, allows

$$V_m \leq 57 \text{ V } (k = 1.2), \quad V_m \leq 69 \text{ V } (k = 1.0),$$

and a 90 V limit allows 66 and 79 V. A nominal submodule voltage of **60 V** ($N = 20$ for 1200 V) gives a peak of 83 V even at $k = 1.2$ and is the recommended design point; 75 V is only acceptable with a tightly regulated submodule voltage ($k \approx 1.0$, peak 86 V). The price of 60 V is 0.09 % inverter efficiency (99.44 instead of 99.53 %, 845 instead of 704 W at 150 kW) and 25 % more transistors and drivers; the gate bump is the same at both voltages.

Table 4: Submodule voltage options for the 1200 V, 150 kW SPB (two EPC2361 per switch, 50 kHz)

V_m (nominal)	48 V	60 V	75 V
Submodules N / transistors / drivers	25 / 300 / 75	20 / 240 / 60	16 / 192 / 48
Peak v_{DS} , $k = 1.0 / 1.2$	59 / 69 V	71 / 83 V	86 / 101 V
Voltage utilization V_m/V_{rated}	48 %	60 %	75 %
Power per transistor ($M = 1.15$)	488 W	610 W	762 W
Switching loss per leg	1.04 W	1.41 W	1.91 W
Inverter efficiency / loss	99.32 % / 1023 W	99.44 % / 845 W	99.53 % / 704 W
High-side gate bump (0 V off)	0.96 V	0.98 V	0.99 V

7 Recommendations

1. Do not reduce the current to fix the bump; it has no effect.
2. Provide a -1 V off-state bias for both gate drives (bipolar auxiliary supply, or a capacitor–Zener level shifter at the driver outputs; check the LT8418 supply limits), or at least place footprints for it, together with do-not-populate gate–source capacitor pads at every gate, so that the measured bump can be corrected on the prototype.
3. Re-evaluate the submodule voltage: with an overshoot of about 11 V, 75 V nominal leaves no margin at the maximum battery voltage. Choose $V_m \approx 60$ V ($N = 20$) unless the submodule voltage is regulated tightly, and repeat the overshoot check at 160 A.
4. Verify on hardware with a double-pulse test and a differential probe on the high-side gate (as on the EPC90156), at room and at operating temperature.

Files: simulation/bb_spice/BB_bump.cir, BB_bump_vth.cir, BB_bump.inc; scripts/gate_bump_study.py; EPC90156 guide and schematic in Datasheet/.